

CIE – 2		
Ramaiah Institute of Technology (Autonomous Institute, Affiliated to VTU)		
Programme: B.E		
Sem: 2 nd	Course: Design Thinking	Course Code: AECC27
CIE: Test 2	Term: 01.03.24 to 22.06.2024	Sec: A,B,C,D,E,F,G,H,I,J,K,L
Max Marks: 30	Duration: 1Hr	Time: 11.00 AM to 12.00 Noon
Credit: 2:0:0	Test Portions: LP-11 to 21	Date: 15.06.2024

Instructions to students: Answer any two full questions out of the three given questions.

Sl#	Question	Marks	Bloom's Level	COs
1a	What is Ideate Phase of design thinking? Write about the myths associated with creativity, innovation and originality.	5	L2	CO3
1b	Is testing phase mandatory in design thinking? Substantiate your answer with suitable reasoning.	5	L2	CO4
1c	Generate atleast three different engineering solutions to reduce traffic jams in Bengaluru.	5	L3	CO5
2a	Bring out the differences between convergent and divergent thinking by giving suitable examples.	5	L2	CO3
2b	What is the end goal of testing stage in design thinking? How is Testing without end user conducted?	5	L2	CO4
2c	Imagine your team is ideating on how to make your locality cycle friendly. Three team members have given the following suggestions: i) Provide segregated and wider bike lanes ii) Install Traffic light signals that prioritizes pedestrians and cyclists iii) Build bike hire infrastructure Build two ideas for each of the above by plussing.	5	L3	CO5
3a	What is rapid prototyping? State the advantages of using it prototype phase of design thinking?	5	L2	CO3
3b	What are the various stages of a product life cycle? Explain with an example.	5	L2	CO4
3c	Scenario: The production process of a pencil was analyzed using the value analysis technique to reduce cost. Wood and paint were the two most expensive elements in producing the pencil, which shared 37.5% of the pencil's total cost. A round-shaped design for the pencil was suggested instead of the hexagonal-shaped design to reduce the manufacturing time and manufacturing cost. In addition, normal paints were suggested instead of expensive glitter paints, and additional care was required while applying them to wood. With the suggested design changes, the production cost of each pencil was reduced by 25%. Does the above scenario fit in as an example of value engineering? Consider two takeaways of value analysis and discuss.	5	L3	CO5